Fixed Dose Combination of Dapagliflozin, Glimepiride and Metformin ER tablets in Indian patients with Type 2 Diabetes: Phase 3 study results

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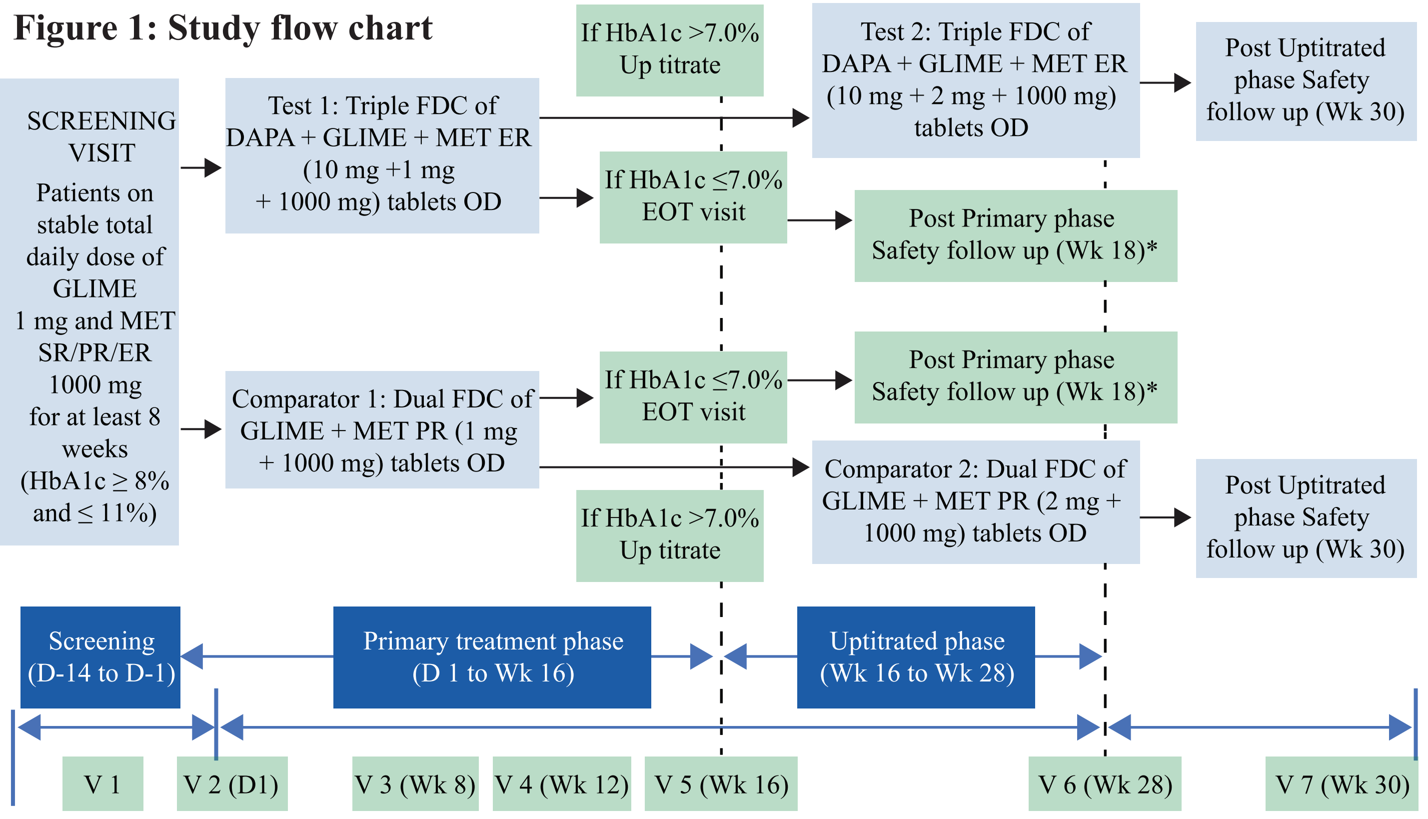
Background and Objectives

- The triple FDC of SGLT2i, SU and Metformin may provide superior glycaemic control, improve therapy compliance, and may have reno-protective function for T2DM.
- The study aimed to assess efficacy and safety of FDC of Dapagliflozin, Glimepiride and Metformin ER tablets in comparison to FDC of Metformin PR and Glimepiride tablets in Indian patients with T2DM.

Materials and methods

- This was a phase III, randomized, open-label, two-arm, multi-centeric, parallel-group, active controlled comparative study (CTRI/2022/03/041424).

Figure 1: Study flow chart



*For patients achieving HbA1c ≤ 7.0%, EOT assessments will be done at Wk 16 and EOS at Week 18.
D: Day; DAPA: Dapagliflozin; EOT: End of Treatment; ER: Extended Release; FDC: Fixed Dose Combination; GLIME: Glimepiride; HbA1c: Glycated Haemoglobin; MET: Metformin; OD: Once Daily; PR: Prolonged Release; SR: Sustained Release; V: Visit; Wk: Week.

- The **primary endpoint** was change in HbA1c from Baseline to Week 16 and **secondary endpoints** included improvement in other glycemic parameters (like FBG and PPBG), proportion of patients achieving HbA1c <7.0% and safety profile.

Results

Demographics Summary:

- In the phase 3 clinical study, 395 patients were enrolled (Test 1 = 195 and Comparator 1 = 200) and at Week 16, 200 patients (Test 2 = 80 and Comparator 2 = 120) entered the Uptitrated phase (12 weeks).
- Of 395 patients, 52.4% patients were males and 47.6% patients were females.
- At Baseline, the mean age of 47.48 ± 10.02 years and mean BMI was 25.90 ± 3.94 kg/m².
- All demographic and baseline characteristics were comparable between Test and Comparator arms.

Mean Change in HbA1c (%) from Baseline to End of Week 16

- Change in HbA1c was statistically significant from Baseline to Week 12 and Week 16 in Test 1 vs. Comparator 1 [p<0.0001 and p=0.0047, respectively] (Figure 2).

Figure 2: Mean change in HbA1c (%) from Baseline to Week 12 and Week 16

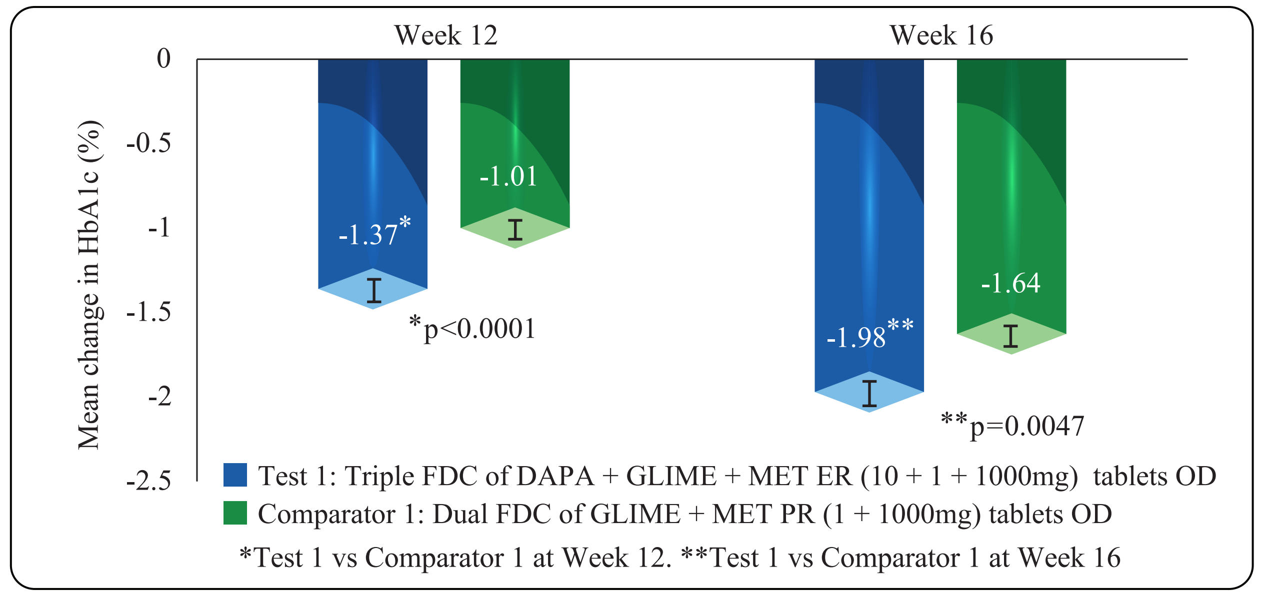
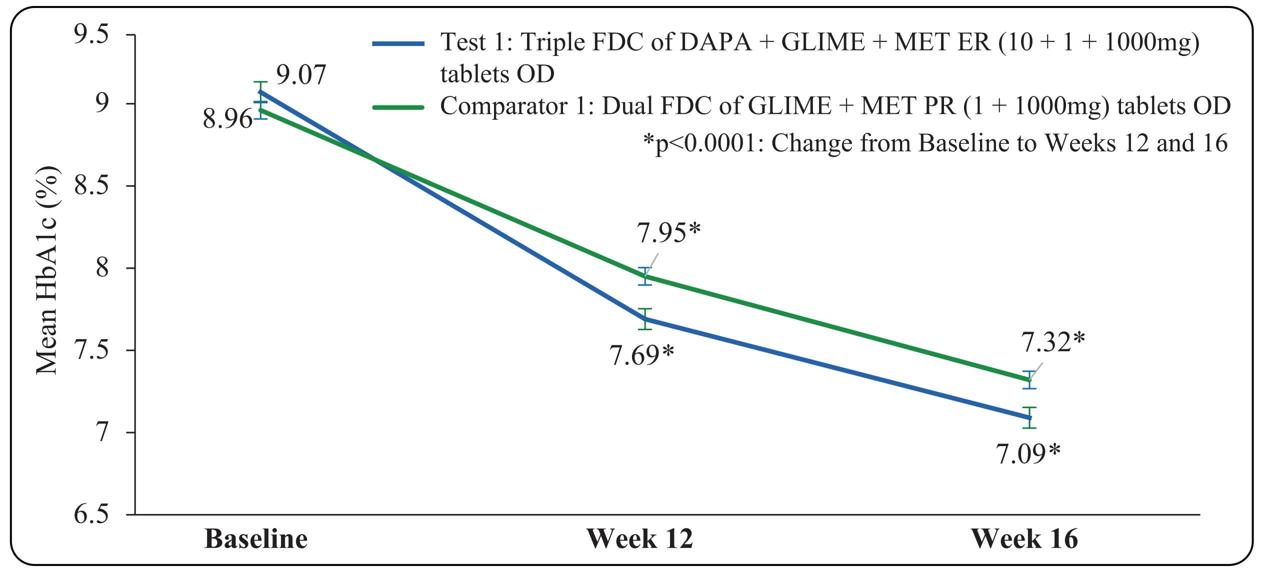


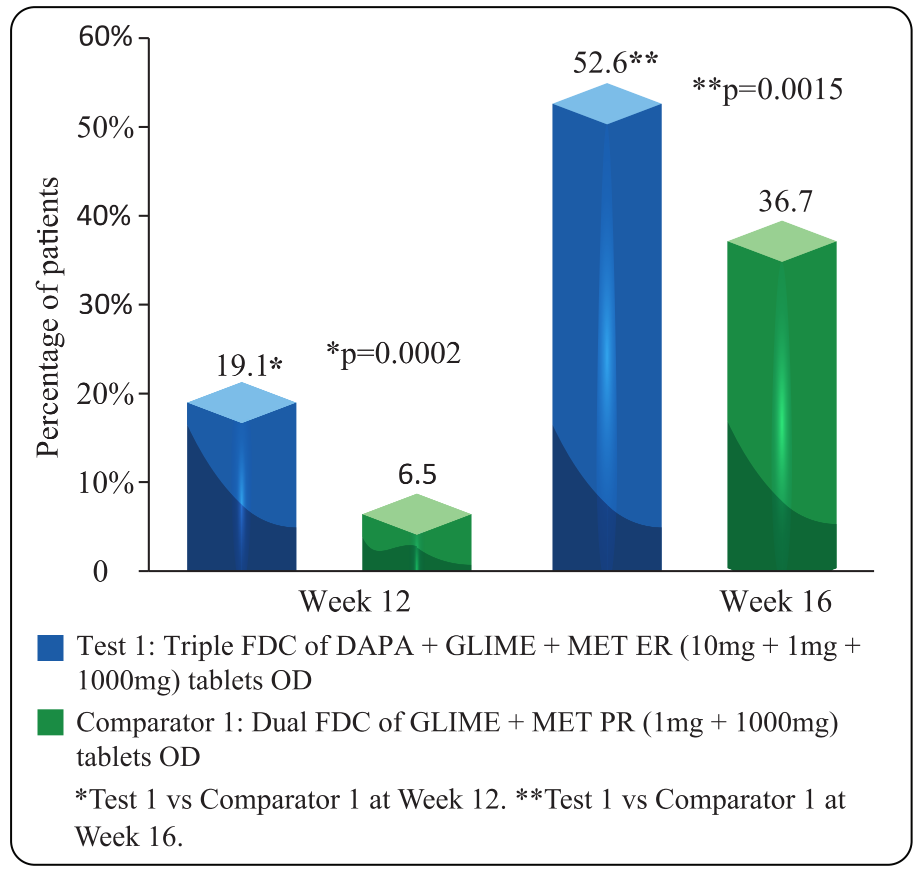
Figure 3: Mean HbA1c (%) at Baseline, Week 12 and Week 16



Change in FBG and PPBG from Baseline at Weeks 16 and 28

- Mean FBG was 171.5±33.23 mg/dL in Test and 166.6±33.08 mg/dL in Comparator at Baseline. Mean PPBG was 252.9±52.77 mg/dL in Test and 246.6±47.48 mg/dL in Comparator at Baseline.
- Reduction in FBG was comparable between Test 1 vs Comparator 1: -45.3±37.30 mg/dL vs -39.4±35.34 mg/dL; p=0.1559 at Week 16.
- Reduction in PPBG was comparable between Test 1 vs Comparator 1: -71.7±56.83 mg/dL vs -60.9±57.90 mg/dL; p=0.2859 at Week 16.
- Statistically significant reduction in FBG and PPBG from Baseline to Week 16 (p<0.0001 each) and Week 28 (p<0.0001 each) were observed within Test and Comparator arms.

Figure 4: Proportion of Patients Achieving HbA1c< 7.0% at Weeks 12 and 16



Change in HbA1c from Baseline at Week 28

- As few patients in the Test arm required uptitrated dose, it led to disproportionate distribution of patients in Test and Comparator arms. Hence, though numerically higher reduction from baseline in mean HbA1c was seen in Test 2 (-2.08%±1.06%) compared to Comparator 2 (-1.80%±1.07%), the results were comparable between Test 2 and Comparator 2 arms at Week 28 (p=0.4943). Reduction in mean HbA1c from baseline was significant in both arms (p<0.0001).

Proportion of Patients Achieving HbA1c <7.0% at Week 28

- Proportion of patients achieving HbA1c <7.0% at Week 28 in Test 2 was 40.5% and in Comparator 2 was 39.5%.

Safety

Primary treatment phase:

- Total 32 TEAEs in 17 (4.3%) patients were reported: 15 events in 9 (4.6%) patients in Test and 17 events in 8 (4.0%) patients in Comparator.
- The most commonly reported TEAEs was pyrexia, headache, and vomiting.
- Event pyrexia in Test and event arthralgia in Comparator were probable/likely' related to study drugs. Rest 30 events were unlikely related to study drugs. None of the events required dose modification.
- One patient in Test arm 1 experienced level 1 hypoglycaemia and did not require active management or dose modification. 4 patients in Test 1 and 2 patients in Comparator 1 required rescue medication.

Uptitrated treatment phase:

- Total 9 TEAEs in 7 (3.5%) patients were reported: 3 events in 3 (3.8%) patients in Test and 6 events in 4 (3.3%) patients in Comparator.
- All events were unlikely related to study drugs and did not require dose modification.
- None of the patients experienced hypoglycaemia or required rescue medication.

Overall safety in both period

- All events were mild in nature
- No serious or severe TEAEs, or TEAEs leading to permanent discontinuation.
- All events were resolved.

Discussion & Conclusion

- ADA and IDF recommends the use of combination drug therapy of SGLT2i+ SU in patients with T2DM inadequately controlled on Metformin.[1,2]
- Triple FDC of DAPA + GLIME+ MET ER showed statistically significant reduction in HbA1c and achieving glycemic control (HbA1c < 7.0%) at Week 12 and Week 16 compared to dual FDC of GLIME+ MET PR, demonstrating better glycemic control.
- All the study treatments were found to be well tolerated.
- As demonstrated in this Phase 3 study, triple FDC showed superior efficacy and comparable safety profile to dual FDC and can be considered as useful therapeutic option in patients with T2DM requiring tighter glycemic control and better therapy compliance.

References:

1. ElSayed NA et al on behalf of ADA. Diabetes Care. 2023; 46(Suppl 1):S5-S9; 2. Federation, I.D., 2019. IDF Clinical Practice Recommendations for managing Type 2 Diabetes in primary Care 2017 [11].

Abbreviations: ADA; American Diabetes Association; BMI: Body Mass Index; DAPA: Dapagliflozin; ER: Extended Release; FBG: Fasting Blood Glucose; FDC: Fixed Dose Combination; GLIME: Glimepiride; IDF; International Diabetes Federation; HbA1c: Glycated haemoglobin or Hemoglobin A1C; MET: Metformin; PPBG: Postprandial Blood Glucose; PR: Prolonged Release; SGLT2i: Sodium-Glucose Transport Protein 2 inhibitors; SU: Sulphonylurea; T2DM: Type 2 Diabetes Mellitus; TEAE: Treatment Emergent Adverse Events.

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